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Human photoplethysmogram: new insight into chaotic characteristics

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ABSTRACT

The photoplethysmogram is widely used in medical settings and sports equipment to measure biological signals. The photoplethysmogram, which is measured noninvasively, can provide valuable information about cardiovascular system performance. The present study sought to investigate the underlying dynamics of photoplethysmographic signals from healthy young human subjects. In previous studies the photoplethysmogram was claimed to be driven by deterministic chaos [Tsuda 1992, Sumida 2000]; however, the methods applied for chaos detection were noise sensitive and inconclusive. Therefore, to reach a consistent conclusion it is important to employ additional nonlinear time series analysis tools that can test different features of the signal's underlying dynamics. In this paper, methods of nonlinear time series analysis, including time delay embedding, largest Lyapunov exponent, deterministic nonlinear prediction, Poincaré section, the Wayland test and method of surrogate data were applied to photoplethysmogram time series to identify the unique characteristics of the photoplethysmogram as a dynamical system. Results demonstrated that photoplethysmogram dynamics is consistent with the definition of chaotic movement, and its chaotic properties showed some similarity to Rossler's single band chaos with induced dynamical noise. Additionally it was found that deterministic nonlinear prediction, Poincaré section and the Wayland test can reveal important characteristics of photoplethysmographic signals that will be important tools for theoretical and applied studies on the photoplethysmogram.

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1. Introduction

Physiological signals derived from the cardiovascular system show an extreme intricacy that arises from the interaction of many processes, structure units and feedback loops in humans. Attempts to improve our understanding of physiological complexity and develop new tools for promising applications for human mental and physical health monitoring have made physiological signals such as electrocardiogram (ECG), electro-encephalogram (EEG), blood pressure, heart rate variability (HRV) and photoplethysmograph (PPG) the subject of recent studies [1–9]. Due to an increase in the successful use of nonlinear time series analysis (NTSA) methods in many scientific disciplines to quantify the complexity of signals [4,9], methods of nonlinear dynamics analysis have become a new and powerful tool for physiological signal investigation. NTSA allows one not only to quantify, but to qualify data.

Many studies have investigated ECG, EEG and HRV signals obtained from healthy human subjects, as well as from

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patients with mental or heart illnesses [1,3–9]. In early studies PPG as well as ECG and HRV were claimed to be chaotic mostly based on results of time-delay reconstructed trajectory, correlation dimension and largest Lyapunov exponent. Later, with the development of methods of nonlinear time series analysis for real world data, evidence for the chaotic nature of many biological signals was questioned [8,9]. Many tools that were previously thought to provide explicit evidence of chaotic motion were discovered to be noise sensitive and could produce misleading results. Thus it is still quite controversial whether these signals' dynamics involve chaotic motion or not [8,9]. Signals like HRV and ECG that were believed to be driven by deterministic chaos have been subjected to detailed reinvestigation [8,9]. Similarly PPG signal was claimed to be chaotic in the early 90s, however nowadays past evidence of chaos was found to be necessary rather than sufficient, especially in biological studies [8,9]. Therefore it is also still not known whether the PPG signal, which is commonly measured by commercially available medical and sport equipment devices to obtain HRV data, oxygen saturation and blood pressure, is chaotic or not. In addition, many of its characteristics are not yet well studied.

The microcirculation of the skin is rather complex dynamic system which is important for skin metabolism and temperature regulation and plays an important role in an organism's defense system. As the skin surface is directly accessible, it has become a valuable organ for many studies. Photoplethysmography (PPG) is one of the widely-used techniques that allows registration of pulsatile changes in the dermal vasculature [10,11].

Photoplethysmography is a simple and low-cost optical technique that can be used to detect blood volume changes in the microvascular bed of tissue. The PPG wave form comprises a pulsatile physiological waveform attributed to cardiac synchronous changes in the blood volume with each heart beat and is superimposed on a slowly varying baseline with various lower frequency components attributed to respiration, sympathetic nervous system activity and thermoregulation [10–12]. Even though pulsation in a finger's capillary vessels (i.e. PPG obtained from finger) in normal subjects was claimed to be chaotic [13,14], only classical methods such as time-delay reconstructed trajectory, power spectrum, correlation dimension (CD) and Lyapunov exponent (LE) and surrogation have been applied to characterize PPG time series [5,13,14]. Positive Lyapunov exponent was believed to provide strong evidence of chaotic behavior. However, these tests (CD and LE) are inconclusive since as it was found in recent studies that they may indicate chaos even in systems that are not driven by chaos [9,15]. Therefore, a clear answer regarding the nature of the PPG signal dynamics cannot be obtained by only applying these types of classical measurements, although they may provide useful results for medical applications.

In this study we applied the time delay embedding method, calculated the power spectrum, largest Lyapunov exponent (LLE), deterministic nonlinear prediction's (DNP) correlation coefficient (CC) and relative route mean square error (RRMSE), Poincaré section, Wayland test translation error and applied method of surrogate data to investigate whether the underlying dynamics of the PPG signal involve motion on a strange attractor and to study the chaotic motion characteristics of the PPG signal. This expanded toolkit is designed to cover most of the important characteristics of chaotic motion and is expected to help us investigate a wider range of PPG signal characteristics, compared with previous studies, and thus to extract its underlying properties. Additionally, in an effort to analyze the PPG signal not only quantitatively, but qualitatively we conducted a comparative analysis of PPG signal and Rosser's single band chaos.

2. Methods and materials

2.1. Photoplethysmography

PPG can be defined as the continuous recording of the light intensity scattered from a given source by the tissues and collected by a suitable photodetector [10]. Modern PPG sensors usually utilize low cost semiconductor technology with LED and matched photodetector devices working at the near infrared (NIR) wavelengths (NIR band 0.8–1 μm), which allows measurement of deep-tissue blood flow [11,12]. The light from the LED is absorbed by hemoglobin, and the backscattered radiation is then detected and recorded. The backscattered light depends on the amount of hemoglobin in the skin, and the obtained result therefore reflects the cutaneous blood flow [10,11].

The microcirculation of the skin is rather complex and dynamic system which is important for skin metabolism and temperature regulation and is an important part of the organism's defense system against invaders. The cutaneous blood supply is carried out into microcirculatory bed composed of three segments—arterioles, arterial and venous capillaries, and venules; most of this microvasculature is contained in the papillary dermis 1–2 μm below the epidermal surface [10–12]. PPG waveform reflects heartbeat synchronized cutaneous blood flow pulsatile changes in the dermal vasculature. The shape of the waveform is related to anacrotic and catacrotic phases. The anacrotic phase corresponds to the rising edge of the pulse, and the catacrotic phase to the falling edge of the pulse curve. The first one is primarily connected with contraction of the heart and therefore with the systolic phase of cardiac cycle, while the second corresponds to diastolic phase (Fig. 1) and wave reflection from the periphery. A dicrotic notch, connected with wave reflected from the periphery, usually can be seen in the catacrotic phase of subjects with healthy arteries [6,11].

The pulsatile component of the PPG waveform that relates to cardiac pulsation is usually called the 'AC' (alternative current) component and its fundamental frequency depends on the heart rate and typically varies around 1–1.4 Hz. The AC component is superimposed onto a large 'DC' (direct current) component that depends on the structure of the tissue and the average blood volume of both arterial and venous blood. The DC component varies slowly due to respiration, vasomotor activity, vasoconstrictor waves, thermoregulation and other slow circulatory changes [10–12].

Although the origins of the components of the PPG signal are not fully understood, it is generally accepted that PPG can provide valuable clinical information about the cardiovascular system [11].

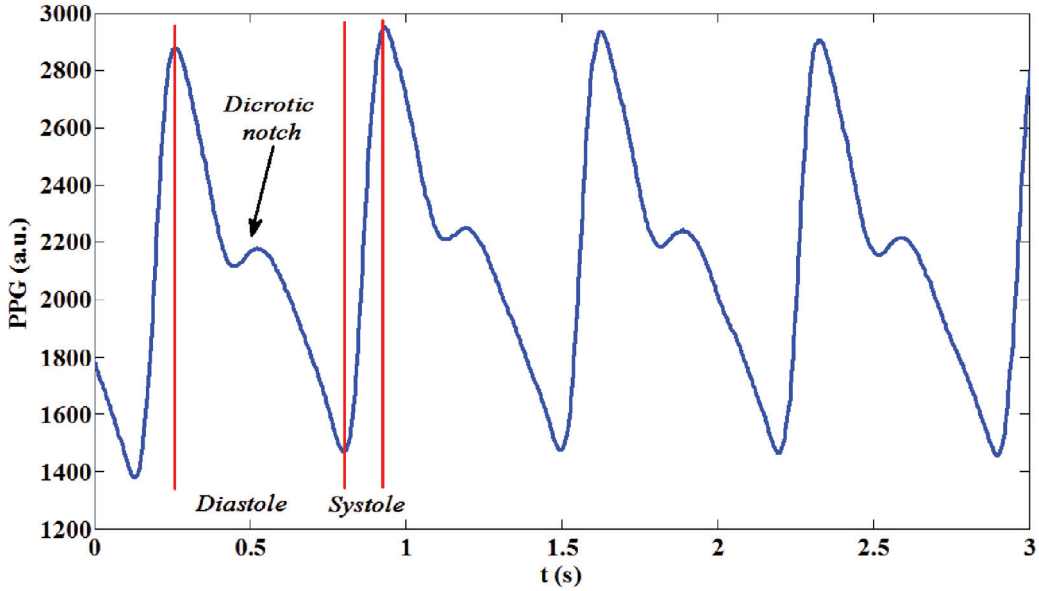


Fig. 1. Components of PPG signal waveform for healthy young subjects.

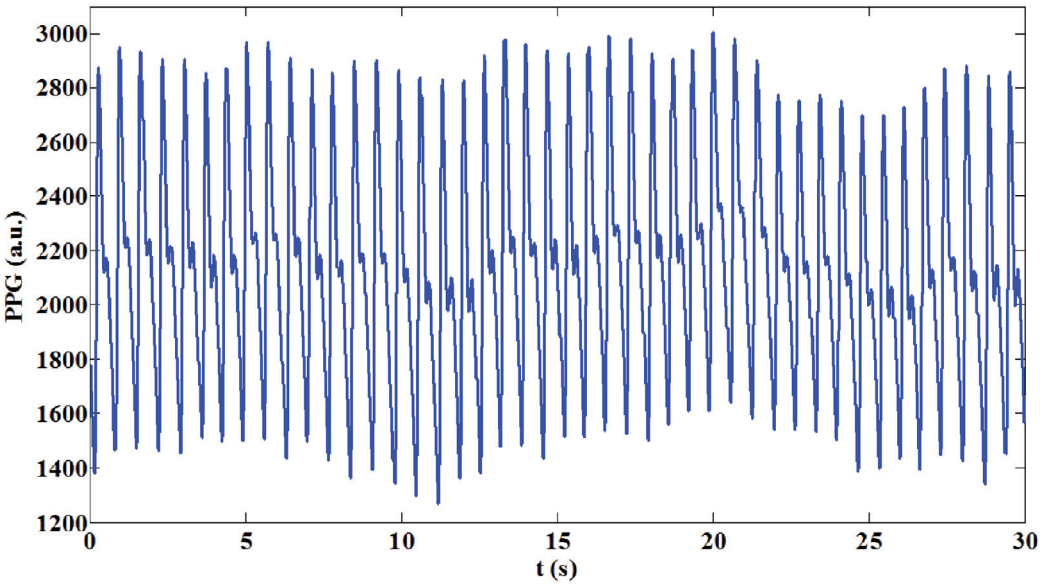


Fig. 2. Example of 30-second long portion of the healthy young subject PPG signal (9th subject's 2nd measurement).

2.2. Data collection

The PPG signal was recorded using a finger PPG recorder by detecting the near infrared light reflected by vascular tissue following illumination with a LED. Data were collected from nine healthy 19–27-year old volunteers among Tokyo University of Agriculture and Technology (TUAT) students. Experimental data collection was approved by TUAT authorities. Written informed consent was obtained from participants prior to the experiment. At the time of the study all subjects were healthy non-smokers, physically active to similar levels, were not taking any medication, and none declared a history of heart disease.

For each subject five measurement repeats were done. The measured period was 5 min with 5 ms sampling steps. For all data collection sessions, a BACS (Computer convenience, Inc.) transmission-mode PPG sensor was located on the right forefinger. Every measurement was preceded by a blood pressure check. Since physical and mental activity, as well as external effects such as temperature, noise etc. can considerably affect cutaneous blood flow [10,11] all measurements were done with the subject in a relaxed sitting position in a room

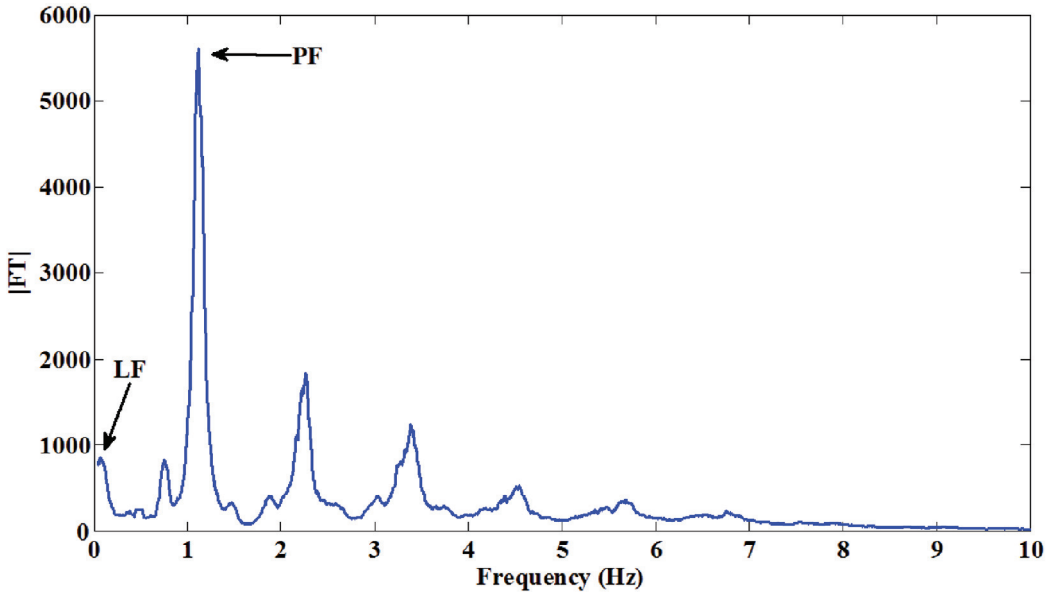


Fig. 3. Example of typical spectra obtained by Fourier analysis, where LF is low frequency and PF is predominant frequency component.

with temperature, noise and vibration control. Each test subject was asked to rest for 5 min under quiet conditions in the laboratory room in a sitting position in which the recordings were obtained, and with the test site uncovered. An example of a 30-second long portion of the obtained PPG signal is shown in Fig. 2.

3. Results

Various definitions of chaos can be found in the literature; Thompson et al. [20] provide quite broad, so called "positive" definition of chaos as "recurrent motion in simple systems or low-dimensional behavior that has some random aspects as well as certain order", which covers wide range of systems that produce chaos and yet have significantly different properties, as for example chaotic Lorenz and Rossler systems.

In this paper to analyze data sets obtained in the experiments described in Section 2.2 and to investigate whether the PPG signal is consistent with the above definition by Thompson et al. and study its chaotic characteristics in details, we applied a complex of nonlinear time series analysis tools. Orbital instability was tested with LLE, determinism with DNP, WTE, recurrence of motion with time-delay embedding and Poincaré section; additionally we applied phase randomized surrogation.

3.1. Spectral analysis

An example of typical plot of the Fourier spectrum in the studied time series is shown in Fig. 3. In Fig. 3, small fluctuations, which indicate environmental noise, can be distinguished around the predominant component (PF) whose period is approximately equal to the heart cycle period. Lower frequency (LF) components correspond to respiration and other effects, such as thermoregulation and nervous system activity. Table 1 shows values of amplitude and frequency corresponding to the predominant component obtained by Fourier transform. As seen from Table 1, all predominant frequencies (PF) are in the range 1.02–1.52 Hz, which is the range of normal heart beat frequencies.

Table 1
Amplitude (|FT|) and frequency (PF) of PPG predominant component obtained by Fourier analysis.

Subject	Repeat									
	1		2		3		4		5	
	|FT|	PF	|FT|	PF	|FT|	PF	|FT|	PF	|FT|	PF
1	5339.0	1.20	5608.5	1.12	5057.4	1.09	4185.4	1.04	1814.5	1.02
2	1877.5	1.24	2063.1	1.21	2906.8	1.21	2276.4	1.12	1839.4	1.22
3	6198.9	1.16	6960.6	1.15	5438.9	1.11	5765.3	1.03	2947.3	1.19
4	1057.7	1.40	3475.7	1.28	2384.0	1.24	4018.1	1.29	3792.7	1.34
5	7843.4	1.07	6379.4	1.09	7048.2	1.05	4945.1	1.03	2784.6	1.05
6	4382.0	1.17	5915.2	1.19	7298.9	1.12	5763.3	1.14	4100.3	1.15
7	7085.3	1.17	6758.1	1.12	5449.0	1.15	4378.1	1.07	4711.1	1.13
8	3028.2	1.09	2557.6	1.10	2451.6	1.11	3563.0	1.12	3006.1	1.06
9	5324.7	1.52	7062.0	1.47	8937.0	1.43	5444.2	1.41	3658.1	1.39

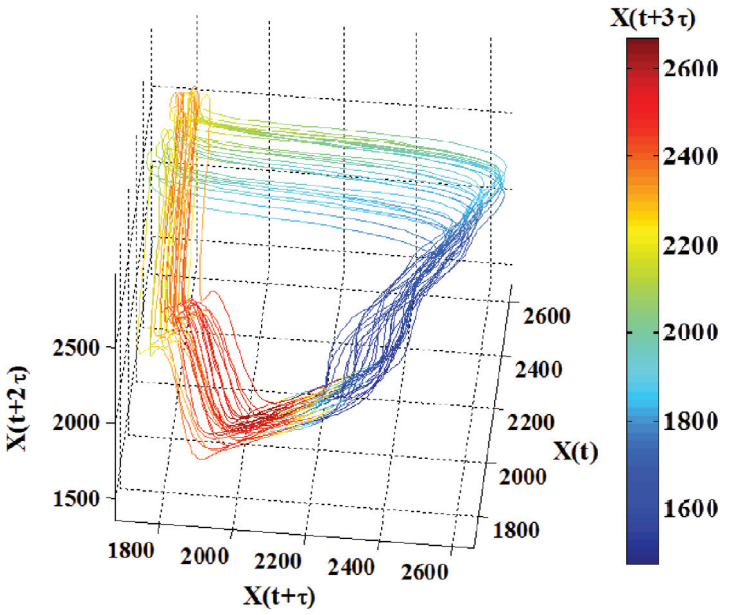


Fig. 4. Trajectory reconstructed in 4 dimensional phase space by time-delay embedding.

3.2. Time delay embedding

By using the time delay embedding technique the possible dynamics of time series can be reconstructed in phase space. The structure of reconstructed trajectory is considered to be an important characteristic of time series; the obtained geometric pattern of the trajectory may provide valuable information about the PPG signal properties, for example, it can reflect the level of physical or mental activity or the level of maturity [13,14].

For time delay reconstruction, the time delay lag τ needs to be sufficiently large to ensure that the resulting individual coordinates are relatively independent; however, it should not be too large to make it completely independent statistically [16]. In this study we have defined the time lag for further calculations as a quarter of the period of the predominant component [17] obtained by Fourier analysis (Table 1).

Following previous studies [7,13,14] where it was shown that at least 4 dimensions need to be used for the PPG signal, we performed a time-delay embedding technique with 4 dimensions to obtain the reconstructed trajectory in phase space. An example (corresponding to the 1st subject's 1st measurement) of our typical data for a time-delay reconstruction is shown in Fig. 4, where the fourth dimension is represented by color. As seen in Fig. 4, the reconstructed trajectory has clear, screwing structure similar to Möbius band.

3.3. Largest Lyapunov exponent

Sensitive dependence to initial conditions is one of the most important properties of the chaotic system. Any chaotic system should have at least one positive Lyapunov exponent, with the magnitude reflecting the time scale on which system dynamics become unpredictable [18,19]. Lyapunov exponent, which provides a qualitative and quantitative characterization of dynamical behavior, is a useful dynamical diagnostic for chaotic systems since it gives a measure of the rate of divergence of neighboring trajectories [18,19].

Table 2
Largest Lyapunov exponents for all collected PPG time series (calculated by Wolf's method).

Subject	Repeat				
	1	2	3	4	5
1	1.041	0.961	1.145	0.744	0.787
2	0.731	0.810	1.099	0.902	0.754
3	1.143	1.565	1.398	1.303	0.968
4	1.100	0.968	0.861	1.019	1.274
5	1.279	1.020	1.209	1.167	0.766
6	1.206	1.692	0.937	1.621	1.242
7	1.250	1.010	1.121	1.157	0.929
8	1.079	1.150	1.194	1.040	1.000
9	1.014	0.938	1.182	1.119	1.008

In this study LLE was calculated with Wolf's method [19], which allows the estimation of LLE from an experimental time series. As many studies have mentioned [8,9,15] most computational methods for estimating LLE have limitations and can produce positive LLE even for non-chaotic systems [9] or can overestimate its value [15].

LLE was calculated for all measured PPG data, and in all cases resulted in a positive LLE as shown in Table 2.

3.4. Deterministic nonlinear prediction

As it is widely known, chaotic systems show predictability in the short term that should decay rapidly, however different systems demonstrate different predictabilities, like for example prediction performance for the Lorenz system in chaotic regime and Rossler's single band chaos differs (as shown in Fig. 5).

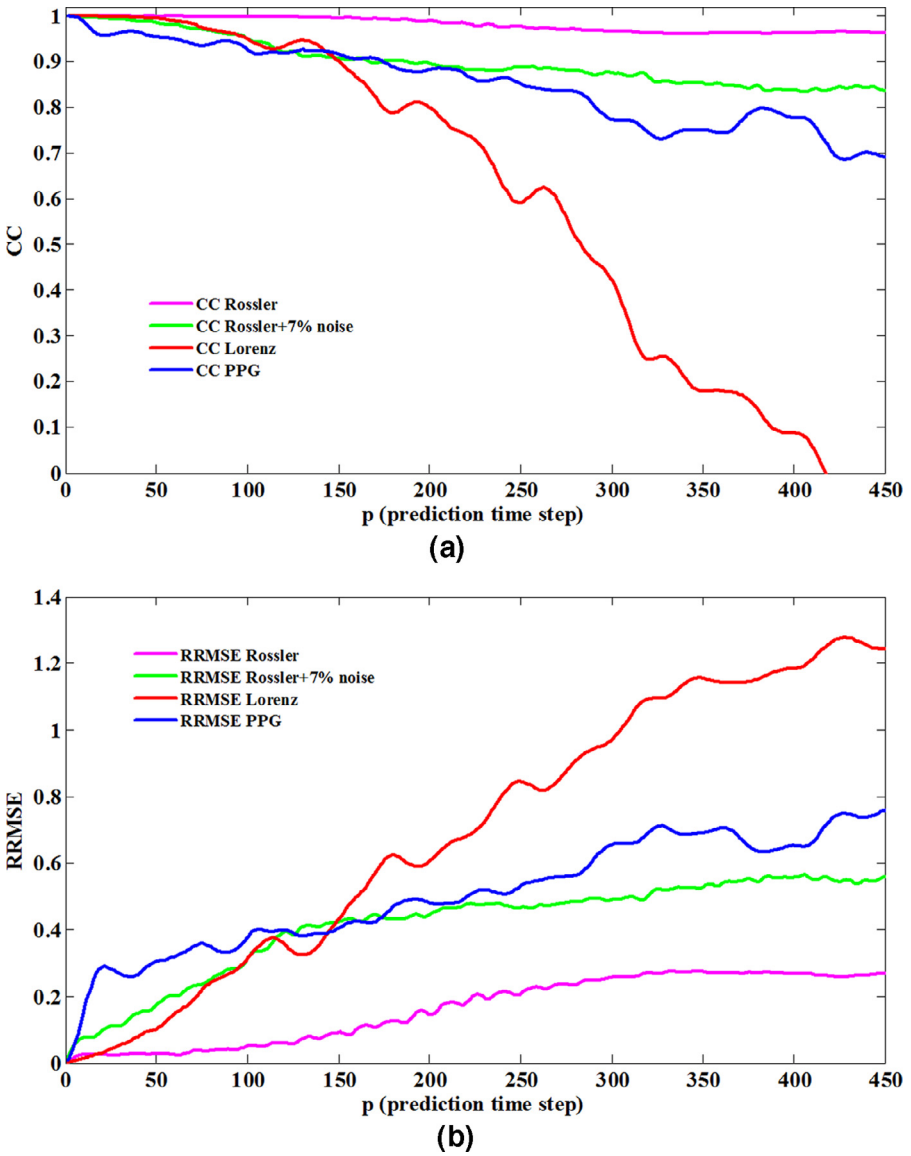


Fig. 5. (a) Correlation coefficient (CC) and (b) relative root mean squared error (RRMSE) curves for nonlinear deterministic prediction of Rossler's single band chaos, Rossler's single band chaos data with 7% additive noise and PPG.

Rossler's system is described by the following equations [20]:

$$\begin{cases} \dot{x} = -y - z, \\ \dot{y} = x + ay, \\ \dot{z} = b + z\,(x - c)\,; \end{cases}$$

and Lorenz system by equations [20]:

$$\begin{cases} \dot{x} = -\sigma\,(x - y)\,, \\ \dot{y} = \rho x - y - xz, \\ \dot{z} = xy - \beta z. \end{cases}$$

For calculation of Rossler's data, the system coefficients chosen were $a = 0.398$, $b = 2$, $c = 4$ which corresponds to single-band chaos [20–22], and for the Lorenz system and $\sigma = 10$, $\rho = 28$, $\beta = 8/3$; numerical simulation was done by the 4th order Runge–Kutta method with time step *0.01*. Information about predictability can be used to distinguish different behaviors. Additionally, the presence of short-term prediction indicates the determinism of the system under investigation [9].

To examine whether PPG time series are predictable in the short-term and investigate how forecasting quality changes with increasing prediction time, we conducted direct deterministic nonlinear prediction (DNP) in this study. Fig. 5(a) shows an example of typical data of CC and Fig. 5(b) RRMSE

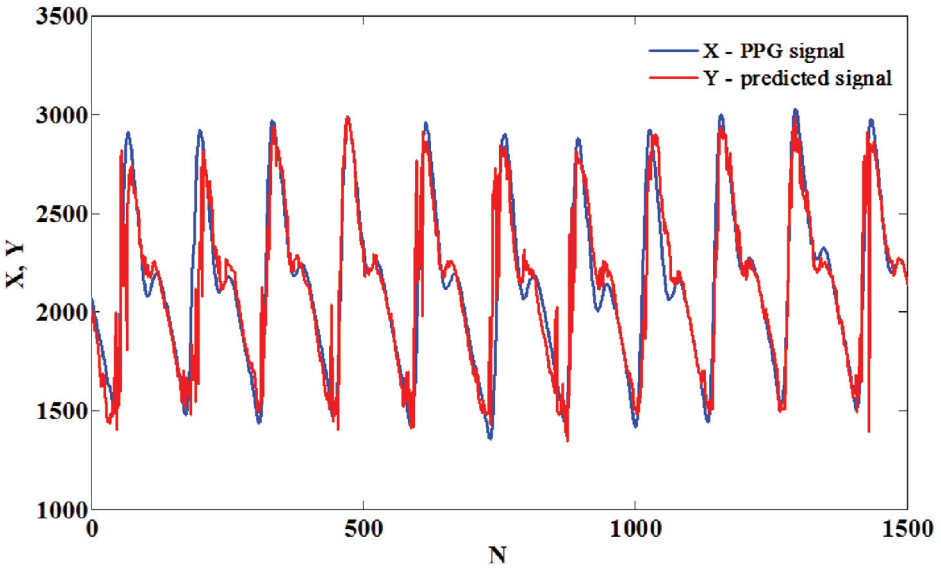


Fig. 6. Performance of deterministic nonlinear prediction. Original signal vs. 160 steps (0.8 s) forward prediction.

between real and predicted signals with increasing prediction time.

While short-time predictability can be easily identified, conclusions regarding long-term prediction are not as straightforward, and in many cases are rather empirical and might be based on researcher experience. For example, prediction performance of well-known chaotic Lorenz data exhibits rapid and apparent decay of CC, which is not always observed for other well-known systems generating chaos. For example, in Fig. 5 Rossler's single band chaos shows high CC (higher than 0.96) over more than 400 steps, while CC corresponding to Lorenz reaches zero. It is important to notice in Fig. 5(a) and (b) that CC and RRMSE curves corresponding to chaotic Rossler band initially demonstrate decay, as would be expected, and later both curves stabilize at a high CC value ($CC > 0.96$) and a low RRMSE value ($RRMSE < 0.28$) over a long period, however, it is known that chaotic systems do not have long-term prediction and therefore despite the stable high CC value and low value of RRMSE these results should not be recognized as a sign of long-term prediction.

As discussed later in Fig. 9, PPG trajectory have some topological similarity to Rossler's single band chaos; therefore, to illustrate changes in the PPG's CC and RRMSE curves it was compared with the CC and RRMSE curves of the corresponding Rossler's data. In addition, since the PPG data were obtained experimentally, it inevitably contains noise, so we also compared PPG prediction results with the same Rossler's system with 7% additive (dynamical) noise; results are shown in Fig. 5(a) and (b).

Fig. 6 shows example of actual vs. 0.8 s (160 time steps) predicted PPG, even after 160 steps forecast still resemble original time series with sufficiently high quality to reproduce not only the general trend of the PPG waveform, but also smaller details such as the dicrotic notch.

As seen from Figs. 5 and 6, time series clearly demonstrated the presence of short-term prediction, which is indicative of underlying determinism. The chosen coefficients for the Rossler system correspond to a chaotic regime, and therefore the obtained Rossler's data should not have long-term prediction. Thus comparison between CC and RRMSE curves corresponding to PPG, Rossler and noise induced Rossler data in Fig. 5 demonstrate the absence of long-term forecasting in the PPG signal, which is typical for chaotic motion. Besides finding that PPG's CC and RRMSE curves showed similar trends with CC and RRMSE corresponding to 7% noise induced Rossler's data, these results have demonstrated that it might be misleading to compare results obtained from PPG with chaotic Lorenz data, which played a role of one of the classical examples of chaos over decades, but actually demonstrates only one of many other possible chaotic behaviors.

In this study DNP became an important method for testing the predictability properties of the PPG signal. In several studies it was claimed that PPG time series of young healthy subjects are deterministic chaos, based mostly on results of the Lyapunov exponent, CD and attractor reconstruction, which as it is nowadays known, may produce misleading results for noise contaminated real-world signals. Results of DNP (Figs. 5 and 6) and its comparison with chaotic Rossler's data provided not only substantial support for the claim of the chaotic nature of PPG, but more importantly, it provides us with quantitative (CC and RRMSE) as well as qualitative (similarity of prediction performance with noise induced chaotic Rossler's data) characteristics of PPG dynamics.

3.5. Poincaré section

The Poincaré section is one of the most powerful tools for qualitative exploration of the dynamics of a system [9], as it enables a demonstration of the process generating chaos in the phase space. The Poincaré section was obtained from a three-dimensional time-delay reconstructed attractor by a clock-wise rotating slicing two-dimensional plane. Fig. 7 demonstrates areas on trajectory sliced by the plane; in Fig. 8, an example of Poincaré sections for the 1st subject's 1st measurement is shown.

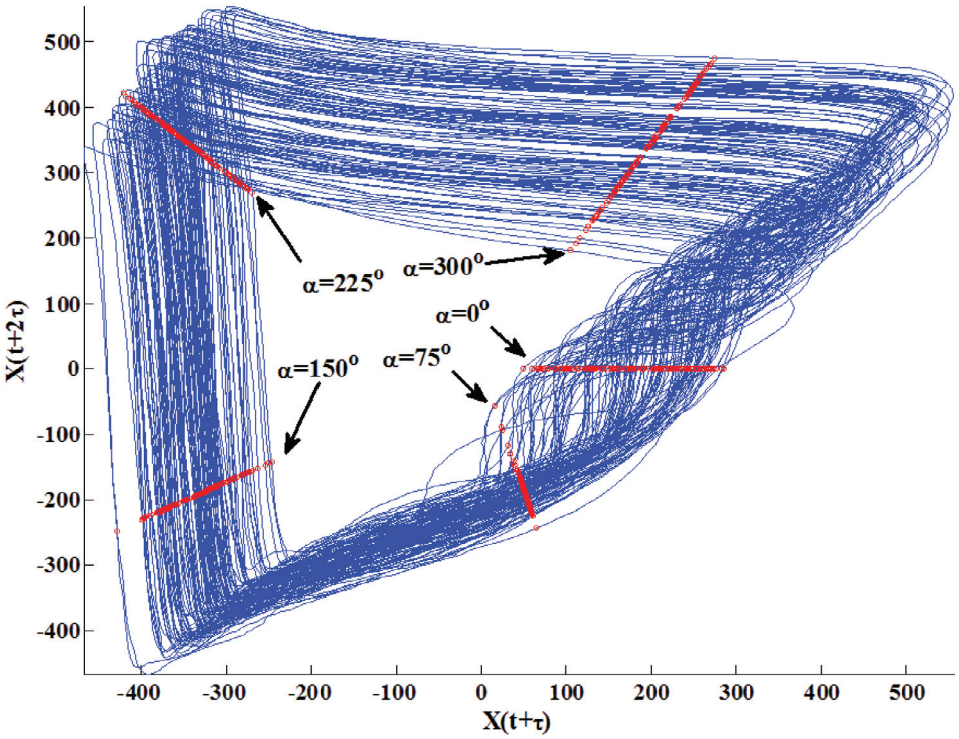


Fig. 7. The areas sliced by the rotating plane on the reconstructed trajectory of PPG signal.

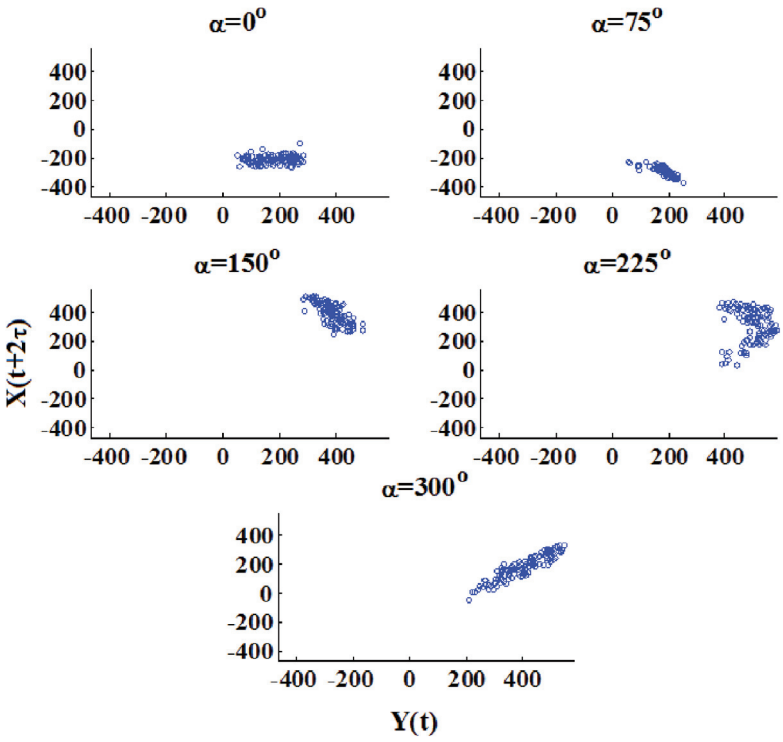


Fig. 8. Poincaré sections for the PPG signal, where $Y(t) = \sqrt{x(t)^2 + x(t+\tau)^2}$.

As seen from Fig. 8 dense trajectories ($\alpha = 75^{\circ}$) tend to expand ($\alpha = 150^{\circ}$), bend ($\alpha = 225^{\circ}$) and stretch ($\alpha = 300^{\circ}$) and then fold back ($\alpha = 0^{\circ}$ and $\alpha = 75^{\circ}$) along the attractor. This stretch-and-fold behavior generates sensitive dependence on the initial conditions, which is recognized as an important property of chaotic dynamics.

3.6. Wayland test

In order to explain the variability observed in complex time series and distinguish whether it is due to external stochastic noise, internal deterministic dynamics, or a combination of both, we applied the Wayland test, which is a computationally simple variation of the Kaplan–Glass method and uses the phase space continuity observed in time series to measure determinism [23].

We calculated Wayland test translation error (WTE), which can quantify smoothness of the flow reconstructed in the phase space. WTE performs well in high levels of uncorrelated noise and provides a robust measure of the determinism in the trajectory reconstructed in the phase space [23]. WTE is insensitive to an overall scaling of the original time series. If the time series is deterministic, then the WTE will be small [23]. For instance, for chaotic Rossler time series (dataset containing 60,000 points) with parameters as described in Section 3.4, the WTE is 0.0002, for Rossler with 10% induced noise the WTE is 0.0045, and for white noise is 1.08 (60,000 points). Table 3 shows the results of WTE calculations for PPG (time series size is 60,000 points).

According to Table 3, most of the WTE values are small, which is an indication of determinism in the PPG data.

Table 3
Results of Wayland test's translation error (WTE) calculation for all collected PPG time series.

Subject	Repeat				
	1	2	3	4	5
1	0.007	0.006	0.005	0.010	0.024
2	0.041	0.041	0.045	0.023	0.048
3	0.023	0.012	0.013	0.015	0.050
4	0.016	0.024	0.020	0.034	0.085
5	0.092	0.024	0.059	0.027	0.017
6	0.015	0.010	0.019	0.019	0.037
7	0.012	0.017	0.016	0.026	0.035
8	0.031	0.030	0.015	0.039	0.048
9	0.018	0.019	0.027	0.045	0.024

3.7. Surrogation

The surrogation approach is to specify a defined null hypothesis and generate a set of surrogate signals that embody a hypothesis about the time series. Then by determining the distribution of the index under investigation obtained from surrogates, empirical statistical boundaries can be found. By comparing the index of the original time series with the distribution of the index from surrogates, the null hypothesis under which surrogates were generated can either be rejected if the value from the original signal does not overlap with the distribution from surrogates, or fail to be rejected if the original index is within distribution with high significance [9,24].

One of the typical applications of surrogate data is to check for determinism and whether data are a realization of a specific random process. In this study the null hypothesis is that the signal is a realization of a linear Gaussian stochastic process. Surrogate time series were created by phase randomization of Fourier transforms for the original PPG data and following inverse Fourier transform. As a result, the obtained surrogates are stochastic, but have the same power spectrum as the original data. Applying this type of surrogates allows one not only to place empirical boundaries, but also to test for nonlinearity of PPG.

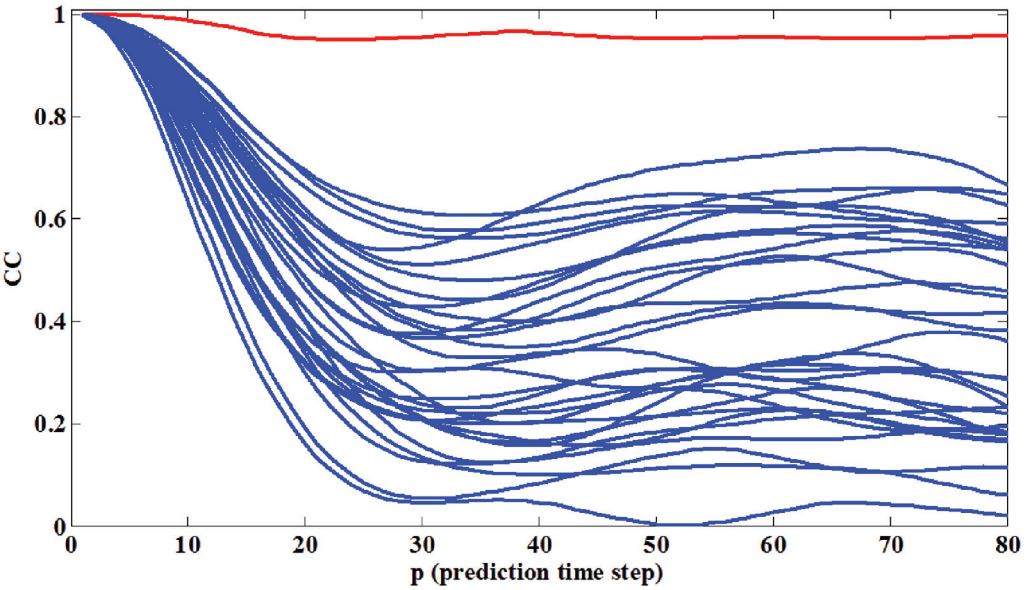


Fig. 9. Deterministic nonlinear prediction correlation coefficient for PPG (red line) and 30 surrogate datasets (blue lines). For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article

DNP were applied to 30 surrogate datasets. Fig. 9 demonstrates the results of DNP for surrogate time series generated under the null hypothesis from the original data. As seen from the CC curve, the original signal is clearly distinct and does not overlap with curves corresponding to surrogates. Since the PPG data and surrogates are clearly distinguishable, the null hypothesis of a linear Gaussian stochastic process should be rejected. Additionally, we may suggest that there is no significant effect of Gaussian noise, which PPG may contain, based on results of DNP.

4. Discussion

The objective of this study was to provide evidence that the PPG signal, obtained from healthy human subjects, is consistent with the definition of chaotic motion given by Thompson et al. [20] and conduct a comprehensive study of the chaotic characteristics of PPG dynamics. Produced by sophisticated mechanisms of the cardiovascular system, the PPG signal is still not fully understood. In the in early 90s PPG was claimed to be chaotic based on time-delay reconstruction, LLE and CD results [13,14], which were previously believed to be sufficient to identify chaos. However, many studies subsequently showed that these measures can be misleading and provide false evidence of chaos in the case of, for example, noise contaminated experimental data [8,9,15]. In addition, it is well known that physiological data are inevitably contaminated by environmental noise and movement artifacts. And the same measures were used in application studies investigating the dependence of PPG characteristics from human subject's mental and physiological conditions [5,13,14].

In an attempt to obtain reliable results, we have applied various methods of NTSA, including classical and widely used ones—power spectrum, time delay embedding, and largest Lyapunov exponent and have conducted deterministic nonlinear prediction, Poincaré section, Wayland test and surrogation, results of which are not only considered to be more reliable in the case of noise-contaminated data, but can also provide new insights into the chaotic properties of PPG dynamics.

Additionally, to study in more detail trajectories' evolving process, we have directly investigated trajectories spreading in phase space. An arbitrary chosen fixed point on an attractor and its nearest neighbor were selected as the starting points of two trajectories. Fig. 10 shows the separation of these two trajectories based on the example of 140 and 300-step long segment for PPG time series (left side) and for Rossler's single band chaos with 7% additive noise (right side), in both cases the trajectories evolve clock-wise. Star and circle markers are corresponding to trajectories originating from an arbitrarily chosen fixed point on an attractor and its neighboring trajectory, respectively, where red color indicates the starting and black color the ending points of the segment. The distance between trajectories increases immediately with increasing evolution time step, however later trajectories are getting closer and remain in the bounded region. Obtained PPG trajectories are continuously bending due to that folding is realized. In this sense PPG's attractor seems to belong to the same folded band topological group as Rossler's single band chaotic attractor, rather than the Lorenz group, in which folding is produced by splitting and layering [20].

The investigated data primarily demonstrated the presence of an experimental attractor with clear structure, which is necessary for a deterministic process. As shown, the time-delay reconstructed PPG attractor had some topological similarity to the attractor obtained from Rossler's single band chaotic data, in which folding is achieved by continuous bending [20]. All of the PPG time series had positive LLE that along with Poincaré section reflects the sensitive dependence on initial conditions. As shown in Figs. 5 and 6 all of the data had short-term predictability. The CC curve tends to decline slowly with an increase of the prediction step. To reach a conclusion on the existence of long-term predictability, results of DNP for PPG were compared with DNP for chaotic Rossler's

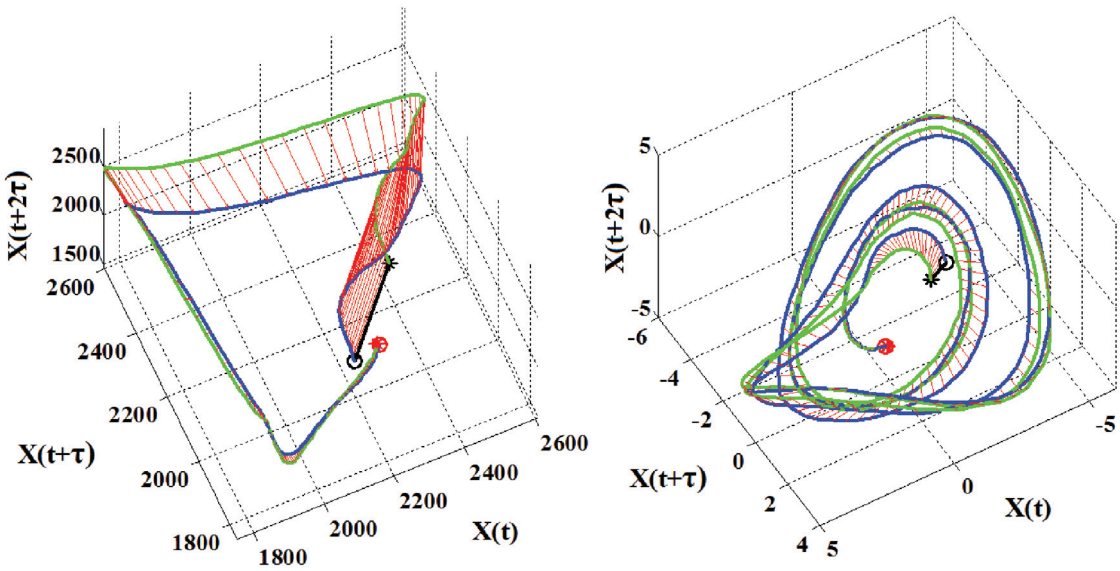


Fig. 10. Trajectory tracings for PPG data (left) and Rossler's single band chaos data with 7% additive noise (right).

data. Results of the comparison explicitly illustrated that PPG data should not have long-term predictability and that prediction performance of the PPG signal is similar to noise induced chaotic Rossler data. The method of surrogate data applied to DNP confirmed the obtained results and demonstrated that there is no significant effect of Gaussian noise that may be contained in the signal on the results of DNP.

To test determinism of the PPG underlying process we applied the Wayland test. Most of the data had a small WTE that clearly indicates strong determinism of the process generating PPG data.

These results are consistent with the so-called "positive" definition of chaos by Thompson et al. [20] as "recurrent motion in simple systems or low-dimensional behavior that has some random aspects as well as certain order". Therefore healthy, young human subjects' PPG dynamics could be concluded as chaotic motion, with chaotic characteristics, such as bending structure of the attractor, predictability properties, which are similar to noise induced Rossler's single-band chaos.

5. Conclusion

In this study we conducted nonlinear time series analysis to investigate the nature of finger PPG data obtained from healthy, young human subjects under resting conditions in a controlled environment. Spectral analysis, time-delay embedding, nonlinear deterministic prediction, Poincaré section calculation, Wayland test and the method of surrogate data were applied to the data to obtain evidence that the studied PPG data are consistent with the definition of chaotic motion. Applied methods aimed to test the chaotic characteristics of PPG dynamics, such as dependence on initial conditions, predictability and determinism. The obtained results provided strong evidence that studied PPG signal dynamics is consistent with chaotic motion. Additionally, the similarity of PPG predictability properties and attractor's topological structure to Rossler's single band chaos was demonstrated.

These results provide new insights into the PPG signal characteristics and demonstrate the usefulness of deterministic nonlinear prediction, Poincaré section and the Wayland test in extracting the underlying chaotic characteristics of the PPG signal. The authors suggest that application of the above mentioned methods of nonlinear time series analysis may contribute to future applied studies on the PPG signal to detect mental and physiological conditions of human subjects.

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References

[1] Ivanov PCh, Nunes Amaral LA, Goldberger AL, et al. Multifractality in human heartbeat dynamics. Nature 1999;399:461–5.
[2] Mascro SA, Asada HH. Photoplethysmograph fingernail sensors for measuring finger forces without haptic obstruction. IEEE Trans Robot Automat 2001;17:698–708.
[3] McClintock PVE, Stefanovska A. Noise and determinism in cardiovascular dynamics. Physica A 2002;314:69–76.
[4] Poon CS, Merrill CK. Decrease of cardiac chaos in congestive heart failure. Nature 1997;389:492–5.
[5] Pham TD, Thang TC, Oyama-Higa M, Sugiyama M. Mental-disorder detection using chaos and nonlinear dynamical analysis of photoplethysmographic signal. Chaos, Solitons & Fractals 2013;51:64–74.
[6] Millasseaua SC, Ritter JM, Takazawa K, Chowienczyk PJ. Contour analysis of the photoplethysmographic pulse measured at the finger. J Hypertens 2006;24:1449–56.
[7] Miao T, Oyama-Higa M, Sato S, Kojima J, Reika S. Chaos of photoplethysmogram in relation to scalp-EEG: a model and experiments. Int J Comput Aided Eng Technol 2012;4(6):557–66.
[8] Glass L. Introduction to controversial topics in nonlinear science: is the normal heart rate chaotic? Chaos 2009;19:1–4.
[9] Shelhamer M. Nonlinear dynamics in physiology: a state-space approach. Singapore: World Scientific; 2007.
[10] Bioengineering of the skin: Cutaneous Blood Flow and Erythema. Edited by E. Berardesca, P. Elsner, H.I. Maibach. CRC Press, USA. 1995.
[11] Allen J. Photoplethysmography and its application in clinical physiological measurement. Physiol Meas 2007;28:1–39.
[12] Tamura T, Maeda Y, Sekine M, Yoshida M. Wearable photoplethysmographic sensors – past and present. Electronics 2014;3:282–302.
[13] Tsuda I. Chaotic pulsation in human capillary vessels and its dependence on mental and physical conditions. Int J Bifurcat Chaos 1992;2(2):313–24.
[14] Sumida T, Arimitu Y. Mental conditions reflected by the chaos of pulsation in the capillary vessels. Int J Bifurcat Chaos 2000;10(9):2245–55.
[15] Nonlinear analysis of Physiological data. Kantz, H., Kurths, J., Mayer-Kress, G. (Eds.). Springer, Berlin, 1998.
[16] Abarbanel HDI. The analysis of observed chaotic data in physical systems. Rev Mod Phys 1993;65:1331–92.
[17] Sakai K. Nonlinear dynamics and chaos in agricultural systems. Netherlands: Elsevier; 2001.
[18] Bezruchko BB, Smirnov DA. Extracting knowledge from time series. Berlin: Springer; 2010.
[19] Wolf A, Swift JB, Swinney HL, Vastano J. Determining Lyapunov exponents from a time series. Physica D: Nonlinear Phenom 1985;16(3):285–317.
[20] Thompson JMT, Stewart HB. Nonlinear dynamics and chaos. U.K: John Wiley and sons; 1991.
[21] Small M, Yu D, Harrison RG. Surrogate test for pseudoperiodic time series data. Phys Rev Lett 2001;87(18):1–4.
[22] Small M, Tse CK, Ikeguchi T. Chaotic dynamics and simulation of Japanese vowel sounds. Proc 2005 Eur Conf Circuit Theory Des 2005;2:169–72.
[23] Wayland R, Bromley D, Pickett D, Passamante A. Recognizing determinism in a time series. Phys Rev Lett 1993;70:580–2.
[24] Theiler J, Eubank S, Longtin A, Galdrikian B, Farmer D. Testing for nonlinearity in time series: the method of surrogate data. Physica D 1992;58:77–94.